

Norway

SLEIPNER

Volve F

F-1

F-1 B

Plan: 15/9-F-1 B Northwest Injector DG2 revp2

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-1	North Reference:	Grid
Wellbore:	F-1 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-1 B Northwest Injector DG2 revp2	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9		
Site Position:		Northing:	6,478,563.53 m
From:	Map	Easting:	435,050.02 m
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in
		Latitude:	58° 26' 29.8068 N
		Longitude:	1° 53' 14.9285 E
		Grid Convergence:	-0.95 °

Well	F-1		
Well Position	+N/-S	3.17 m	Northing:
	+E/-W	-3.53 m	Easting:
Position Uncertainty	0.00 m		Wellhead Depth:
			91.00 m
			Latitude:
			58° 26' 29.9072 N
			Longitude:
			1° 53' 14.7075 E
			Water Depth:
			91.00 m

Wellbore	F-1 B		
Magnetics	Model Name	Sample Date	Declination
			(°)
	MNETICREFERENCE	02.05.2013	-1.20
			Dip Angle
			(°)
			71.62
			Field Strength
			(nT)
			50,512

Design	15/9-F-1 B Northwest Injector DG2 revp2		
Audit Notes:			
Version:	Phase:	PLAN	Tie On Depth:
			2,585.00
Vertical Section:	Depth From (TVD)	+N/-S	+E/-W
	(m)	(m)	(m)
	145.90	3.17	-3.53
			Direction
			(°)
			29.16

Survey Tool Program	Date	14.11.2014		
From	To	Survey (Wellbore)	Tool Name	Description
(m)	(m)			
150.15	210.70	15/9 F1 36" MWD (F-1)	Magnetic, old	Magnetic Tools (MWD, EMS)
210.70	450.00	15/9-F-1 North Upside Pilot DG2 revp1 (F-	Gyro, old	Other Gyro Tools
450.00	650.00	15/9-F-1 North Upside Pilot DG2 revp1 (F-	NOT_useMWD gyro, Baker Hughes' MWD gyro service (GyroTrak)	
650.00	1,355.00	15/9-F-1 North Upside Pilot DG2 revp1 (F-	Magnetic, IFR, mag-corr	Magnetic Tools (MWD, EMS)
1,355.00	2,585.00	15/9-F-1 North Upside Pilot DG2 revp1 (F-	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)
2,585.00	3,088.00	15/9-F-1 B Northwest Injector DG2 revp2 (	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)
3,088.00	3,463.17	15/9-F-1 B Northwest Injector DG2 revp2 (	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)

Planned Survey										
Measured	Inclination	Azimuth	Vertical	+N/-S	+E/-W	Vertical	Dogleg	Build	Turn	
Depth	(°)	(°)	Depth	(m)	(m)	Section	Rate	Rate	Rate	
(m)			(m)			(m)	(°/30m)	(°/30m)	(°/30m)	
2,585.00	30.22	36.85	2,437.81	533.88	174.11	550.00	0.000	0.000	0.000	
2,595.00	30.22	36.85	2,446.45	537.90	177.13	554.99	0.000	0.000	0.000	
2,688.33	25.99	50.77	2,528.85	569.68	207.10	597.34	2.500	-1.358	4.476	
3,080.70	25.99	50.77	2,881.54	678.41	340.28	757.19	0.000	0.000	0.000	
3,332.57	5.00	50.79	3,122.90	720.73	392.12	819.40	2.500	-2.500	0.003	

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Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
3,412.57	5.00	50.79	3,202.60	725.14	397.52	825.88	0.000	0.000	0.000	
3,463.17	5.00	50.79	3,253.00	727.92	400.94	829.98	0.000	0.000	-0.002	

Design Targets										
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude		Longitude
- hit/miss target										
- Shape										
F-1 B NW Injector Base	0.00	360.00	3,122.90	720.73	392.12	6,479,284.00	435,442.00	58° 26' 53.3058 N		1° 53' 38.3636 E
- plan hits target center										
- Point										

Casing Points							Casing Diameter (in)	Hole Diameter (in)
Measured Depth (m)	Vertical Depth (m)	Name						
220.90	220.90	30"					30.000	36.000
1,355.00	1,343.22	20"					20.000	26.000
2,585.00	2,437.81	13 3/8"					13.375	17.500
3,088.00	2,888.12	9 5/8"					9.625	12.250
3,412.60	3,202.63	7"					7.000	8.500

Checked By: _____	Approved By: _____	Date: _____
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